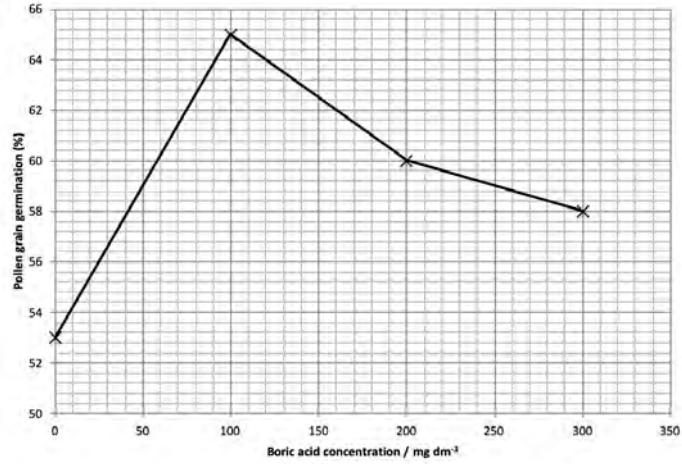


| Question Number | Answer                                                                            | Additional Guidance                              | Mark |
|-----------------|-----------------------------------------------------------------------------------|--------------------------------------------------|------|
| 3a              | <ul style="list-style-type: none"> <li>number of grains germinated (1)</li> </ul> | Accept percentage germination (of pollen grains) | (1)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                              | Additional Guidance | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| 3b              | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>because it might affect {(pollen grain) germination / enzymes (involved in germination)} (1)</li> <li>it is (thought to be) the optimum for {(pollen grain) germination / enzymes (involved in pollen germination)} (1)</li> </ul> |                     | (2)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                  | Additional Guidance                                                                                                          | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------|
| 3ci             | <p>A graph with the following features:</p> <ul style="list-style-type: none"> <li>• A axes correctly orientated (1)</li> <li>• L1 axes correctly labelled (1)</li> <li>• L2 axes with correct units (1)</li> <li>• P correct plotting on a suitable linear scale for both axes (1)</li> <li>• R points joined with ruled straight lines (1)</li> </ul> |  <p>max 4 for bar chart as loses mp R</p> | (5)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                           | Additional Guidance                                                      | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------|
| 3cii            | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>• (all concentrations of) boric acid enhance germination (1)</li> <li>• germination is higher in 100 mg dm<sup>-3</sup> than all other concentrations / germination shows optimum value at 100 mg dm<sup>-3</sup> (1)</li> </ul> | <p>Accept optimum between 0 and 200</p> <p>Allow consequential error</p> | (2)  |

| Question Number | Answer                                                                                                                                                                            | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3di             | <p>A calculation that includes:</p> <ul style="list-style-type: none"> <li>• calculation of increase in rate</li> <li>• calculation of percentage increase in rate (1)</li> </ul> | <p>Accept values from 211 to 214 for 0 at 2 hrs</p> <p>e.g. rate at 0 is <math>212 \div 2 = 106</math><br/> difference = <math>\{130.5 - 106 / 24.5\}</math></p> <p><math>(24.5 \div 106) \times 100 = 23</math> (%) ignore decimal places</p> <p><math>130.5 - 106 \div 130.5 = 18.8</math> gets 1</p> <p><b>alternatives</b></p> <p>1. rate at 0 is <math>211 \div 2 = 105.5</math><br/> = <math>\{130.5 - 105.5 / 25\}</math></p> <p><math>(25 \div 105.5) \times 100 = 23</math> (%) ignore decimal places</p> <p>2. rate at 0 is <math>213 \div 2 = 106.5</math><br/> = <math>\{130.5 - 106.5 / 24\}</math></p> <p><math>(24 \div 106.5) \times 100 = 22</math> (%) ignore decimal places</p> <p>3. rate at 0 is <math>214 \div 2 = 107</math><br/> = <math>\{130.5 - 107 / 23.5\}</math></p> <p><math>(23.5 \div 107) \times 100 = 22</math> (%) ignore decimal places</p> <p><math>130.5 - 106 \div 130.5 = 18.8</math> gets 1</p> <p><math>130.5 - 105.5 \div 130.5 = 19.2</math> gets 1</p> <p><math>130.5 - 106.5 \div 130.5 = 18.3</math> gets 1</p> <p><math>130.5 - 107 \div 130.5 = 18</math> gets 1</p> <p>Correct answer with no working gains 2 marks (2)</p> | (2)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Additional Guidance                                                                                                                                                                      | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3dii            | <p>An answer that includes five of the following points:</p> <ul style="list-style-type: none"> <li>• make (100, 200, 300 mg dm<sup>-3</sup>) boric acid solutions from 500 mg dm<sup>-3</sup> solution by dilution (1)</li> <li>• control variable identified (1)</li> <li>• view through high power of microscope (1)</li> <li>• suitable method for measuring pollen tube length described (1)</li> <li>• record results every hour for 4 hours (1)</li> </ul> | <p>Ignore serial</p> <p>e.g. pH, sucrose concentration, temperature, light</p> <p>Accept magnification quoted <math>\geq 100\times</math></p> <p>e.g. stage micrometer and graticule</p> | (4)  |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                      | Additional Guidance | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| 3(e)            | <p>An answer that includes the following:</p> <ul style="list-style-type: none"> <li>• {both / germination <b>and</b> growth} show optimum at 100 mg dm<sup>-3</sup> (1)</li> <li>• for growth boric acid concentrations above 100 mg dm<sup>-3</sup> are inhibitory whereas for germination all concentrations above 100 mg dm<sup>-3</sup> are stimulatory (1)</li> </ul> |                     | (2)  |